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Project Review: Modules 1 and 2

Code-Mixed Retrieval-Augmented Generation: An Empirical Study of Chunking, Embedding and Query Normalisation for Tamil–English (Tanglish) Queries

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Abstract

Millions of Tamil speakers type their questions in *Tanglish*: Tamil words written in the Latin alphabet, freely mixed with English ("scholarship ku last date enna?"). Retrieval-Augmented Generation (RAG) systems, which answer questions by first retrieving relevant documents and then generating an answer from them, are almost always built and tested on English queries. This project asks a simple, measurable question: *what happens to each component of a RAG retrieval pipeline when the query is Tanglish instead of English or Tamil script?*

Modules 1 and 2 define the problem, survey related work, design the system, and implement and run a baseline retrieval pipeline. Using the public nlpctx/tamil-english-retrieval dataset, we built a 500-document Tamil-script news corpus with 2,154 queries, each available in four forms: English, Tamil script, romanised Tamil and code-mixed Tanglish. We evaluated five retrieval configurations (BM25 on words, BM25 on character 3-grams, dense retrieval with multilingual-e5-small, and two Reciprocal Rank Fusion hybrids) and two query-normalisation modes that transliterate romanised Tamil into Tamil script (the off-the-shelf ITRANS scheme and a small set of Tanglish rules written for this project). All experiments ran on a 2-core CPU with no GPU.

The results are clear. Tamil-script queries are retrieved well (MRR@10 up to 0.923). Romanised Tamil queries collapse: dense retrieval reaches only MRR@10 = 0.143 (Recall@10 = 0.294), and the best raw configuration only 0.178. Transliterating them to Tamil script with the Tanglish rules raises character-3-gram BM25 to MRR@10 = 0.684 and Recall@10 = 0.812 ($p < 0.001$ under both query-level and document-cluster bootstrap tests) at a cost of about 0.3 ms per query, and the rules beat the ITRANS scheme on every retriever. Code-mixed queries are handled mainly by the dense model (MRR@10 = 0.687), and normalisation does not improve dense retrieval for them ($p = 0.45$). Equal-weight BM25 + dense fusion, a common default in RAG tutorials, *hurts* cross-script retrieval (English queries on Tamil documents: MRR@10 0.788 → 0.710; with a third, character-3-gram list, 0.788 → 0.305). An earlier version of this pipeline overstated that loss (0.788 → 0.187) because zero-score BM25 documents were fed into the fusion; this report gives the corrected numbers. We also found that 22% of the dataset's "code-mixed" queries contain no Tamil word at all, which motivates the human-written Tanglish question set planned for Module 3.

Keywords: Retrieval-Augmented Generation, code-mixing, Tanglish, Tamil, transliteration, BM25, dense retrieval, multilingual embeddings, Reciprocal Rank Fusion.

Chapter 1: Introduction

1.1 Background

A Retrieval-Augmented Generation (RAG) system answers a question in two steps [1]. First, a *retriever* searches a collection of documents and returns the few most likely to contain the answer. Second, a *generator* (a large language model) reads those documents and writes the answer. If the retriever returns the wrong documents, the generator has nothing correct to work with, so retrieval quality sets the ceiling for the whole system.

Retrievers come in two broad families. *Sparse* or lexical retrievers such as BM25 [2] score documents by the query words they share, weighted by how rare those words are. *Dense* retrievers encode the query and each document as vectors with a neural network and rank documents by vector similarity; multilingual dense models such as multilingual-E5 [3], BGE-M3 [4] and LaBSE [5] are trained so that text with the same meaning lands close together across languages. Many RAG systems combine both families with Reciprocal Rank Fusion (RRF) [6], which merges the two ranked lists.

1.2 Problem statement

Tamil has tens of millions of speakers in India, Sri Lanka and abroad, and in chat, search and social media many of them do not type in Tamil script. They type romanised Tamil, often mixed with English words: Tanglish. The same question can arrive in at least four forms:

Form	Example
English	Who is responsible for the Israel-Hamas conflict
Tamil script	இஸ்ரேல் ஹமாஸ் மோதலுக்கு யார் பொறுப்பு
Romanised Tamil	Israel Hamas mothalukku yaar poruppu
Code-mixed (Tanglish)	Israel Hamas conflict ku yaar responsible

Table 1.1: One information need in four query forms (English, romanised and code-mixed forms taken from the evaluation dataset).

The documents a Tamil RAG system searches (government notices, news, college circulars) are usually in Tamil script or English. Romanised Tamil shares neither the script of Tamil documents nor the vocabulary of English documents, and romanisation is unstandardised: the same word can be spelt *kurithu*, *kuritthu* or *kuriththu*. Transliteration-based mixed-script retrieval itself has a history at the Forum for Information Retrieval Evaluation (FIRE), where Tamil appeared in word-level language-labelling subtasks [19], [20]. **What is missing, and what this project addresses, is that no prior study evaluates RAG retrieval components (dense, lexical, fusion) for Tamil–English code-mixed and romanised queries.**

1.3 Motivation

1. **Real users.** Students, farmers and citizens in Tamil Nadu routinely ask questions in Tanglish. A college help-desk bot or an e-governance assistant that works only for English or Tamil-script queries fails the users it is meant to serve.
2. **Hidden failure.** RAG tutorials recommend hybrid BM25 + dense retrieval as a safe default. Whether that advice holds across scripts has not been checked for Tamil.

3. **Cheap fixes may exist.** If a simple, fast normalisation step recovers most of the lost accuracy, it can be added to any existing RAG system without retraining a model.
4. **Research gap.** Code-mixed NLP for Dravidian languages has focused on sentiment and offensive-language classification [7], [8]. The FIRE transliterated-search and mixed-script IR tracks (2013–2016) ran their retrieval subtasks on Hindi and Bengali, and included Tamil only in word-level language labelling [19], [20]; later code-mixed retrieval tasks used Bengali–English [9], [10], and code-switching retrieval benchmarks used Chinese and Japanese [11]. No prior study evaluates RAG retrieval components (dense, lexical, fusion) for Tamil–English code-mixed and romanised queries.

1.4 Objectives

The overall objective is an empirical study, with a reusable benchmark and open code, of chunking, embedding and query normalisation for Tanglish queries in RAG. The objectives for the full project are:

1. **O1.** Build an evaluation set in which the same information need is expressed as English, Tamil-script, romanised and code-mixed queries over a Tamil document collection. *(Modules 2 and 3.)*
2. **O2.** Measure how far retrieval quality drops for each query form, for lexical, dense and hybrid retrievers. *(Module 2, done.)*
3. **O3.** Design and evaluate lightweight query normalisation (transliteration to Tamil script) and measure the accuracy it recovers and the latency it costs. *(Module 2, done for two methods.)*
4. **O4.** Evaluate Tamil-aware document chunking against standard chunkers on longer documents. *(Module 3.)*
5. **O5.** Measure end-to-end answer quality of a RAG system (retrieval plus LLM generation) for each query form. *(Module 4.)*
6. **O6.** Publish the findings, benchmark and code. *(Module 5.)*

1.5 Scope and limitations

In scope: text queries in four forms; Tamil-script and English documents; BM25, character n-gram BM25, dense multilingual embeddings and RRF fusion; rule-based and scheme-based transliteration; CPU-only execution so the work is reproducible on a student laptop; standard IR metrics with confidence intervals.

Out of scope for Modules 1–2: speech input, neural transliteration models (IndicXlit [22] is planned as a Module 3 baseline), fine-tuning embedding models, re-rankers, and answer generation (Module 4). Chunking is designed in Module 1 but not varied in Module 2, because the documents in the current dataset are short (median 326 characters), so each document is already a single chunk.

Known limitation of the current data: every query in the dataset was generated by an LLM, not written by a person. Section 5.2 shows this matters, and Module 3 addresses it.

1.6 Contributions of Modules 1–2

1. A problem definition and a design for a factorial study of RAG components under four query forms.

2. A literature survey of 19 relevant works, including the FIRE mixed-script IR tracks, with a comparison table and a stated research gap.
3. A working, CPU-only retrieval pipeline that reuses the retriever and metric design of the author's ChunkLab project [12].
4. A Tanglish-to-Tamil-script rule transliterator and a comparison against the ITRANS scheme.
5. Real results for 2,154 queries × 8 query sets × 5 retrievers on a Tamil corpus, plus 4 query sets × 5 retrievers on an English corpus, with query-level and document-cluster bootstrap tests and latency.
6. An audit showing that a public "code-mixed" Tamil query set is substantially English, which shapes the Module 3 data plan.

Chapter 2: Literature Survey

2.1 Retrieval-Augmented Generation and retrievers

Lewis et al. [1] introduced RAG, combining a dense passage retriever with a sequence-to-sequence generator, and showed that retrieval grounds generation in external knowledge. BM25 [2], a probabilistic term-weighting function, remains a strong baseline and is the lexical retriever in most RAG stacks. Cormack et al. [6] proposed Reciprocal Rank Fusion, which sums 1/(k + rank) across ranked lists with k = 60; it needs no score calibration and is the usual way to build a "hybrid" retriever.

2.2 Multilingual embedding models

LaBSE [5] learns language-agnostic sentence embeddings for 109 languages from translation pairs. Multilingual-E5 [3] is a family of text-embedding models (small, base, large) trained with weakly supervised contrastive pre-training followed by supervised fine-tuning; it expects "query:" and "passage:" prefixes. BGE-M3 [4] supports dense, sparse and multi-vector retrieval in one model across more than 100 languages. These models are trained mainly on native-script text, so their behaviour on romanised Tamil is an open empirical question, and one this project measures.

2.3 Code-mixed language processing and retrieval

Chandu et al. [13] crowd-sourced a code-mixed factoid question-answering dataset with 1,694 Hinglish, 2,848 Tamlish (Tamil–English) and 1,391 Tenglish questions. It is the closest earlier Tamil–English resource, but it targets factoid QA, not RAG retrieval components. Chakravarthi et al. [7] built a Tamil–English (Tanglish) sentiment corpus from YouTube comments, and the Dravidian-CodeMix track at FIRE 2020 [8] (continued in 2021) used it for sentiment-classification shared tasks. These works show that Tanglish is common online, but they are classification tasks.

Mixed-script IR at FIRE. Gupta et al. [21] formalised mixed-script information retrieval, where queries and documents in one language appear in both the native and the Roman script. They estimated from Bing logs that 6% of unique queries from India contained romanised Hindi words, and they expanded queries with cross-script term equivalents learned by a deep autoencoder, raising MRR from 0.686 to 0.874 on Hindi song-lyrics retrieval (FIRE 2013 collection, 62,888 documents, 25 queries). The FIRE Transliterated Search track began in 2013 with language labelling and back-transliteration for Hindi, Bengali and Gujarati plus Hindi lyrics retrieval; in 2014 the labelling subtask added Tamil, Kannada and Malayalam [19]. Renamed Mixed Script IR in 2015, it labelled words in code-mixed text for eight languages including Tamil, while its retrieval subtask remained Hindi only (song lyrics, movie reviews and astrology; 25 test queries) [20]. One 2015 system transliterated every Roman-script word to Devanagari and indexed character 2- to 6-grams with vowel signs removed, which gave the best nDCG@10 among its runs [20]; this is a direct precedent for our transliteration plus character-n-gram design. The 2016 edition moved to Bengali–English question classification and Hindi–English tweet retrieval, with a best MAP of only 0.0377 [19]. In short, transliteration-based mixed-script IR is an established FIRE topic, but its retrieval tasks never covered Tamil and predate dense retrievers and RAG.

Transliteration resources. Aksharantar [22] is a 26-million-pair transliteration corpus for 21 Indic languages, and IndicXlit, trained on it, is an open neural model for romanised-to-native transliteration; it is the natural neural baseline for our rule-based normaliser. IndicCMix [23] is a recent AI4Bharat dataset with about 104,800 Tamil rows, each giving an English sentence with a native-script code-mixed version, a full Tamil translation and a casual romanised version; it has no accompanying paper or retrieval relevance judgements.

The FIRE Code-Mixed Information Retrieval (CMIR) track [9], [10] is the closest recent retrieval benchmark. Its 2024 edition used 50 Bengali–English code-mixed queries over 107,900 social-media comments; the best system, built on GPT-3.5 Turbo, reached MAP 0.70 [9]. The 2025 edition added native-script Bengali [10]. The language is Bengali, the documents are social-media comments, and RAG components are not studied.

Zeng et al. [11] built a human-annotated code-switching retrieval benchmark (CSR-L) and an 11-task CS-MTEB suite for Chinese and Japanese mixed with English. They found that code-switching degrades statistical, dense and late-interaction retrievers, with drops of up to 27%, and that vocabulary expansion does not close the gap. This supports our hypothesis but does not cover Tamil or romanised script.

2.4 Cross-lingual retrieval and RAG component studies in Indian and low-resource settings

Balasubramaniyam and Madushanka [14] compared query translation with cross-lingual embeddings (LaBSE, multilingual-E5, BGE-M3) for Sinhala and Tamil queries over English e-government content, using 500 human-verified question–answer pairs. BGE-M3 reached Recall@15 of 95.6% for Tamil–English. Their Tamil queries are in native script only; romanised and code-mixed queries are not studied.

Singh [15] ran a controlled factorial study of 3 parsers × 3 chunking strategies × 5 dense embedding models, plus BM25, on 800 questions over Indian government regulatory documents, with mixed-effects models and bootstrap intervals. It is the methodological model for this project, but it is English-only.

Chhoun et al. [16] compared four chunking strategies (recursive, Khmer-aware, sentence-based, LLM-based) for Khmer agricultural RAG with BGE-M3, and found 300-character recursive chunking best. It motivates our Tamil-aware chunker in Module 3, although it used only 18 question–answer pairs.

Madhusudhanan G. [17] studied indirect prompt injection and reasoning monitorability across English, Tamil and Tanglish in a small case study of one model. It is evidence that Tanglish is now studied in LLM safety, a possible later direction for this work.

2.5 Datasets

The nlpctx/tamil-english-retrieval dataset [18] contains 9,299 LLM-generated query rows over 2,183 Tamil documents from the Mozhi Tamil corpus (IndicCorpV2 news text), with IndicTrans2 English translations. Each row has an English, a Tamil-script, a romanised and a code-mixed query, plus an answer snippet. It is released under Apache-2.0 and has no accompanying paper or reported baselines; to our knowledge, Chapter 5 gives the first retrieval baselines on it.

2.6 Summary of the literature

#	Work	Year	Languages / script	Task	Method / models	Key result	Gap relative to this project
1	Lewis et al. [1]	2020	English	Open-domain QA	DPR + BART (RAG)	Retrieval grounding improves QA	English only
2	Cormack et al. [6]	2009	English	Rank fusion	RRF, k = 60	Beats individual rankers	Not tested across scripts
3	Feng et al. (LaBSE) [5]	2022	109 languages	Sentence embedding	Dual encoder	Strong bitext retrieval	Native scripts only
4	Wang et al. (mE5) [3]	2024	~100 languages	Text embedding	Contrastive pre-training	Strong multilingual retrieval	Romanised Tamil untested
5	Chen et al. (BGE-M3) [4]	2024	100+ languages	Text embedding	Dense + sparse + multi-vector	Strong multilingual retrieval	Romanised Tamil untested
6	Chandu et al. [13]	2018	Hinglish, Tamlish, Tenglish	Factoid QA	Crowd-sourced data	2,848 Tamlish questions	QA, not RAG components
7	Chakravarthi et al. [7], [8]	2020	Tamil–English	Sentiment classification	Corpus + shared task	Tanglish corpus	Classification, not retrieval
8	Chanda and Pal; Chanda et al. [9], [10]	2024–25	Bengali–English	Code-mixed IR	GPT-3.5, SBERT, BM25	Best MAP 0.70 (2024)	Bengali; social media; no RAG study
9	Zeng et al. [11]	2026	Chinese / Japanese + English	Code-switching IR	Sparse, dense, late interaction	Drops up to 27%	Not Tamil; not romanised
10	Balasubramaniyam and Madushanka [14]	2026	Sinhala, Tamil → English	Cross-lingual IR	Translation vs LaBSE / mE5 / BGE-M3	BGE-M3 Recall@15 95.6% (Tamil)	Native script only
11	Singh [15]	2026	English	RAG component study	Parser × chunker × embedder	Components interact	English only
12	Chhoun et al. [16]	2026	Khmer	RAG chunking	4 chunkers + BGE-M3	300-char recursive best	Khmer; 18 QA pairs
13	Madhusudhanan G. [17]	2026	English, Tamil, Tanglish	LLM safety	Case study, 1 model	Descriptive only	Safety, not retrieval
14	nlpctx dataset [18]	2026	Tamil, English, romanised, code-mixed	Retrieval data	LLM-generated queries	No baselines published	Synthetic queries; no study
15	Gupta et al. [21]	2014	Hindi (Devanagari + Roman)	Mixed-script IR	Autoencoder term equivalents	MRR 0.686 → 0.874	Hindi lyrics; pre-dense, no RAG

#	Work	Year	Languages / script	Task	Method / models	Key result	Gap relative to this project
16	FIRE MSIR 2015 [20]	2015	8 Indic incl. Tamil (labelling); Hindi (retrieval)	Language labelling; mixed-script retrieval	Transliteration + char n-grams	Best NDCG@10 0.716	Tamil only in labelling
17	FIRE MSIR 2013–16 [19]	2016	Hindi, Bengali (+ Tamil labelling, 2014)	Mixed-script IR	Transliteration, query expansion	Best tweet MAP 0.0377 (2016)	No Tamil retrieval
18	Madhani et al. (Aksharantar) [22]	2023	21 Indic languages	Transliteration	IndicXlit transformer	26M word pairs	Not evaluated for retrieval
19	IndicCMix [23]	2026	11 Indic incl. Tamil	Parallel code-mixed text	Translated + romanised renderings	~104.8k Tamil rows	No relevance judgements

Table 2.1: Literature survey summary.

2.7 Research gap

Transliteration-based mixed-script IR has a history at FIRE [19]–[21], and transliteration plus character-n-gram indexing was already used there [20]; we do not claim that idea as new. What the surveyed work never combines is: (a) **Tamil–English code-mixed and romanised queries**, (b) **retrieval for RAG with modern dense, lexical and fusion components**, and (c) **a controlled comparison of those components** on the same information needs. The FIRE retrieval subtasks were Hindi or Bengali [19], [20]; native-script Tamil cross-lingual retrieval has been studied [14]; code-mixed retrieval has been studied for Bengali [9] and for Chinese and Japanese [11]; RAG components have been compared for English [15] and Khmer [16]. No prior study evaluates RAG retrieval components (dense, lexical, fusion) for Tamil–English code-mixed and romanised queries; this project fills that intersection.

Chapter 3: System Analysis and Design

3.1 Existing system

A typical RAG deployment for an Indian institution today works as follows. Documents are split into fixed-size chunks, embedded with an English or multilingual embedding model, and stored in a vector index. Optionally a BM25 index is added and fused with RRF. The user's query is passed *as typed* to both retrievers.

Drawbacks of the existing approach:

1. **No script handling.** A romanised Tamil query shares no characters with Tamil-script documents, so BM25 cannot match anything, and the embedding model sees an unfamiliar romanised string.
2. **Hybrid fusion assumed safe.** RRF gives the lexical list equal weight even when it matches only incidental tokens, which can push correct dense results down.
3. **Chunkers ignore Tamil.** Character or token chunkers can split Tamil words inside a grapheme cluster, and sentence splitters are tuned for English punctuation.
4. **Evaluated only in English.** Retrieval quality is rarely measured for the query forms real users type.

3.2 Proposed system

The proposed system adds a script-aware front end and a measurement harness to a standard RAG retriever:

1. **Script and language identification per token.** Each Latin-script token is labelled English or romanised Tamil, using English word frequencies (the wordfreq library) plus a list of about 60 common Tanglish particles and function words (*ku, la, enna, pathi, panni, yaar* and so on) that would otherwise be mistaken for English.
2. **Query normalisation.** Romanised Tamil tokens are transliterated into Tamil script; English tokens are kept unchanged. Two transliterators are compared: the ITRANS scheme from the indic-transliteration library, and a Tanglish rule set written for this project.
3. **Multiple retrievers.** BM25 on whole words, BM25 on character 3-grams (robust to spelling variation), and dense retrieval with multilingual-e5-small.
4. **Fusion.** RRF over two lists (BM25 + dense) or three lists (BM25 + character-3-gram BM25 + dense). A BM25 list contains only documents with a positive score.
5. **Evaluation harness.** Recall@k, MRR@10, nDCG@10, bootstrap confidence intervals, paired significance tests and per-stage latency, run for every query form.
6. **Later modules.** A Tamil-aware chunker (Module 3) and an LLM generator with answer-quality evaluation (Module 4).

3.3 Existing versus proposed system

Aspect	Existing RAG pipeline	Proposed system
Query forms	Typed text, treated as one language	English, Tamil script, romanised and code-mixed, each measured separately
Script handling	None	Token-level language ID + transliteration of romanised Tamil

Aspect	Existing RAG pipeline	Proposed system
Lexical matching	Word BM25 only	Word BM25 + character 3-gram BM25
Dense model	Often English-only	Multilingual (mE5-small now; BGE-M3 and LaBSE in Module 3)
Fusion	RRF applied by default	RRF measured per query form; harm detected and reported
Chunking	Fixed size, script-agnostic	Tamil-aware chunker (Module 3)
Evaluation	English test set, often none	2,154 queries × 4 forms, CIs, significance, latency
Hardware	Often GPU	CPU-only (2 cores), reproducible on a laptop

Table 3.1: Existing versus proposed system.

3.4 System architecture

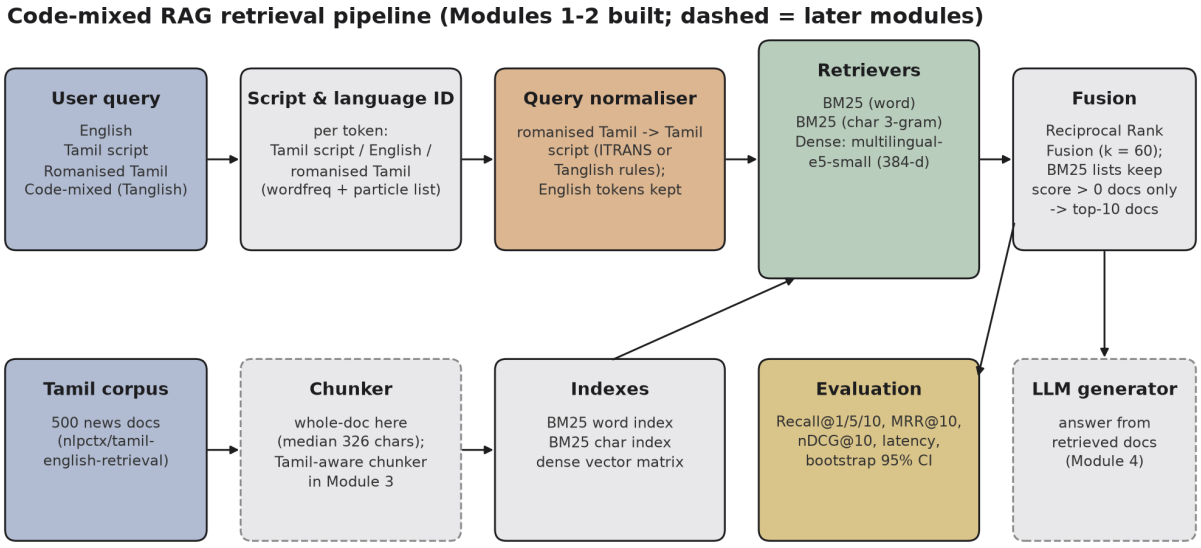


Figure 3.1: Architecture of the proposed code-mixed RAG retrieval pipeline. Solid boxes are built in Modules 1–2; dashed boxes come in later modules.

The query path runs along the top row: the user's query passes through token-level language identification, then the normaliser, then the three retrievers, whose ranked lists are fused by RRF into a top-10 result. The corpus path along the bottom row builds the indexes: documents pass through the chunker (whole documents in Module 2) into a BM25 word index, a BM25 character index and a dense vector matrix. The evaluation block compares the top-10 list with the gold document. The LLM generator is added in Module 4.

3.5 Module description

Module	Content	Status
M1: Problem and design	Problem statement, objectives, literature survey, existing vs proposed system, architecture, methodology	Complete (this report)
M2: Data and baseline retrieval	Dataset subset, query audit, normaliser, 5 retrievers, metrics, results, examples	Complete (this report)
M3: Human Tanglish set and chunking	~500 human-written Tanglish questions; longer documents; Tamil-aware chunker vs ChunkLab chunkers; BGE-M3 and LaBSE	Planned
M4: Generation-side evaluation	LLM generator; answer correctness and faithfulness per query form	Planned

Module	Content	Status
M5: Paper and release	Paper, public benchmark and code	Planned

Table 3.2: Project modules.

3.6 Hardware and software

Hardware used for every Module 2 experiment: 2 virtual CPU cores (ARM64), 7 GB RAM, no GPU.

Software: Python 3.12; PyTorch 2.15.0 (CPU build); sentence-transformers 6.2.0; transformers 5.22.1; rank-bm25 0.2.2; indic-transliteration 2.3.82; wordfreq 3.1.1; pandas 3.1; NumPy 2.5; matplotlib 3.11; huggingface_hub 2.4. The ChunkLab code base [12] supplied the retriever and metric design (BM25, dense, hybrid RRF with k = 60, and the Precision/Recall/MRR/nDCG definitions). Module 2 reimplements these as small standalone functions so the experiment runs without ChunkLab's Streamlit interface.

Chapter 4: Methodology

4.1 Research questions

- **RQ1.** How much does retrieval quality drop when the same information need is expressed in romanised or code-mixed Tamil instead of Tamil script or English?
- **RQ2.** Which retriever type (lexical word, lexical character n-gram, dense, hybrid) is most robust to each query form?
- **RQ3.** How much of the drop does transliteration-based query normalisation recover, at what latency cost, and does a Tanglish-specific rule set beat a generic scheme?
- **RQ4 (Module 3).** Does Tamil-aware chunking improve retrieval on longer documents for each query form?
- **RQ5 (Module 4).** How do retrieval differences carry through to generated-answer quality?

Module 2 answers RQ1 to RQ3.

4.2 Dataset preparation

The source is nlpctx/tamil-english-retrieval [18] (9,299 query rows over 2,183 documents, all news). To keep the study CPU-feasible, 500 documents were sampled uniformly at random with a fixed seed (42), and **every** query row generated for those documents was kept, giving 2,154 queries. Each query has exactly one relevant document, the one it was generated from. Each query exists in all four forms, so every form is evaluated on the same 2,154 information needs, which allows paired comparisons.

Two corpora were built from the same 500 documents: the **Tamil corpus** (original Tamil-script text; the main setting) and the **English corpus** (the dataset's IndicTrans2 English translation; a secondary setting). No chunking was applied: documents have a median length of 326 characters, so each fits easily in the embedding model's 512-token limit.

4.3 Query normalisation

Normalisation applies to romanised and code-mixed queries. Each query is split into runs of Latin letters and other characters. A Latin token is kept as English if its Zipf frequency in English (from wordfreq) is at least 3.0 and it is not in the Tanglish particle list; otherwise it is treated as romanised Tamil and transliterated. Digits and punctuation pass through unchanged.

ITRANS mode applies the ITRANS-to-Tamil mapping of the indic-transliteration library to the lower-cased token. ITRANS is a formal romanisation scheme designed for Sanskrit-family text. For example, it maps *e* to the long vowel ஏ and *n* to the dental ந in every position, which does not match how Tanglish is typed.

Tanglish rules mode is a greedy longest-match transliterator written for this project. Its complete rule set:

Tanglish spelling	Tamil output	Example (actual output of the code)
zh	ழ	kizhakku → கிழக்கு

Tanglish spelling	Tamil output	Example (actual output of the code)
th, dh	த	thadai → தடை
t, d (not followed by h)	ட	tamil → டமில்
tr, rr	ற்ற	payitrapatta → பயிற்றபட்ட
ndr	ன்ற	endru → என்று
nth, ndh	ந்த	vandhu → வந்து
nd, nt	ண்ட (ந்த at the start of a word)	kandu → கண்டு
ng, ngk	ங்க	ingu → இங்கு
nj, nch	ஞ்ச	manjal → மஞ்சல்
ny	ஞ	nyayam → ஞயம்
n at the start of a word	ந	neelam → நீலம்
other n; nn	ன; ன்ன	enna → என்ன
ch, s	ச	seyarkai → செயர்கை
tch	ட்ச	kaatchi → காட்சி
sh; ksh	ஷ; க்ஷ	shankar → ஷன்கர்
k, g, c, q, kh, gh	க	gangai → கங்கை
p, b, f, ph, bh	ப	bharatham → பரதம்
j, z; h; x	ஜ; ஹ; க்ஸ	haridwar → ஹரிட்வர்
m, y, r, l; v, w	ம, ய, ர, ல; வ	vaazhai → வாழை
doubled consonant other than r, n (kk, pp, tt, ll…)	consonant + pulli + consonant	kudikkalama → குடிக்கலம
aa; ee or ii; oo or uu; ai; au; ae	vowel signs, shown on க: கா; கீ; கூ; கை; கௌ; கே	maagaana → மாகான
a, i, u, e, o after a consonant	shown on க: க (inherent a); கி; கு; கெ; கொ	kurithu → குரிது
consonant with no following vowel	consonant + pulli (க்)	yaar → யார்
vowel at the start of a word	independent vowel (அ, ஆ, இ…)	aasiriyar → ஆசிரியர்

Table 4.1: The complete rule set of the Tanglish-to-Tamil transliterator, as implemented in normalise.py (longest match first). Every example is the code's actual output.

The rules are deliberately simple and were written before the evaluation was run, from knowledge of Tamil orthography only; they were not tuned on the test queries. They cannot recover distinctions that Tanglish spelling does not encode (ல/ள typed as *l*, ன/ண/ந as *n*, or த/ட both typed as *t*: *tamil* becomes டமில் not தமிழ், which is why character n-gram matching is also evaluated.

4.4 Retrievers

1. **BM25 (word).** Okapi BM25 with rank-bm25 defaults (k1 = 1.5, b = 0.75). Tokens are maximal runs of Latin letters, digits and Tamil-block characters (U+0B80 to U+0BFF),

lower-cased. Including the whole Tamil block keeps vowel signs attached to their consonants.

2. **BM25 (character 3-gram).** The same BM25 over character 3-grams of each word with boundary markers (#word#). It tolerates spelling variation, which matters for transliteration output.
3. **Dense (multilingual-e5-small).** 118M parameters, 384-dimensional embeddings, with the required "query:" and "passage:" prefixes. Vectors are L2-normalised and ranked by dot product, which equals cosine similarity. This small model was chosen so every experiment runs on a 2-core CPU.
4. **Hybrid-RRF.** Reciprocal Rank Fusion of the top-100 lists from BM25 (word) and dense, with k = 60, as in ChunkLab. rank-bm25 gives every document that shares no term with the query a score of exactly 0; such documents are removed from the BM25 list before fusion, so a query with no lexical match contributes no BM25 list at all.
5. **Hybrid-RRF3.** RRF of BM25 (word), BM25 (character 3-gram) and dense, with the same zero-score filter.

Correction to an earlier run. The first version of this pipeline (and an earlier draft of this report) passed the full top-100 BM25 list to RRF. When a query matched nothing, all scores were 0 and the "ranking" was simply documents 0 to 99 in index order, which RRF then rewarded. This affected 72% of English and 77% of code-mixed queries on the Tamil corpus (no word-BM25 match). Section 5.6 reports the corrected numbers and keeps the old ones, labelled *naive*, for comparison. The single BM25 retrievers are also filtered now, which removes a few chance hits among zero-score documents and lowers some BM25 scores by up to 0.022 Recall@10.

4.5 Evaluation metrics

Each query q has one relevant document; let r(q) be its rank in the top-10 list (undefined if absent). Over the query set:

- **Recall@k** (k = 1, 5, 10): the fraction of queries whose relevant document appears in the top k. With one relevant document per query this equals the hit rate.
- **MRR@10** (Mean Reciprocal Rank): the average of 1 / r(q), counting 0 when the relevant document is not in the top 10.
- **nDCG@10**: the average of 1 / log2(r(q) + 1), counting 0 when absent. With one relevant document the ideal DCG is 1, so no further normalisation is needed.
- **Latency**: wall-clock time per query for each stage (normalisation, BM25 scoring, dense query encoding, dense search, fusion), measured one query at a time on 200 random queries; median and 95th percentile are reported.

Statistical testing. 95% confidence intervals for MRR@10 use 1,000 bootstrap resamples of queries. Differences between two configurations on the same queries use a two-sided paired bootstrap test with 2,000 resamples; with 2,000 resamples, the smallest reportable p-value is below 0.001. Because queries cluster within documents (2,154 queries from 500 documents, about 4.3 per document), every key comparison is repeated with a stricter *cluster* bootstrap that resamples whole documents together with all their queries.

4.6 Experimental protocol

1. Sample 500 documents (seed 42) and keep all 2,154 associated queries.
2. Create the ITRANS and rules versions of each romanised and code-mixed query, giving eight query sets: English, Tamil script, romanised, code-mixed, romanised + ITRANS, romanised + rules, code-mixed + ITRANS, code-mixed + rules.
3. Encode both corpora and all eight query sets with multilingual-e5-small; build the word and character BM25 indexes.
4. Run all five retrievers for all eight query sets on the Tamil corpus (40 runs) and for the four raw query sets on the English corpus (20 runs): 60 main runs. The two naive fusion variants (Section 4.4) are re-run on all 12 query sets for comparison, giving 84 runs and 180,936 ranked lists.
5. Compute metrics, confidence intervals, query-level and document-cluster significance tests and latency; audit query composition; extract example outputs.

Chapter 5: Module 2 Implementation and Results

5.1 Dataset statistics

Property	Value
Source	nlpctx/tamil-english-retrieval (Hugging Face, Apache-2.0)
Full dataset	2,183 documents, 9,299 query rows, all news domain
Documents sampled (seed 42)	500
Queries per form	2,154 (all rows for the sampled documents)
Query forms	4 raw (English, Tamil script, romanised, code-mixed) + 4 normalised = 8 query sets
Relevant documents per query	1 (the source document)
Document length (Tamil, characters)	mean 391, median 326, max 2,795
Corpora	Tamil script (main); English translation (secondary)
Ranked lists evaluated	84 runs (60 main + 24 naive-fusion comparison runs) × 2,154 queries = 180,936

Table 5.1: Evaluation subset.

5.2 Query audit: how code-mixed are the queries?

Before looking at retrieval scores, we examined the queries themselves.

Query set	Avg tokens	Latin-script tokens (%)	Latin tokens judged English (%)	Tamil-script tokens found in corpus vocabulary (%)
English	5.6	98.9	85.1	n/a
Tamil script	5.0	0.3	n/a	77.7
Romanised	5.1	98.9	14.2	0.0
Code-mixed	5.7	98.9	69.0	n/a
Rom -> ITRANS	5.1	14.0	100.0	15.2
Rom -> rules	5.1	13.8	100.0	29.9
CM -> ITRANS	5.7	68.2	100.0	16.6
CM -> rules	5.7	68.2	100.0	24.9

Table 5.2: Token composition of each query set (2,154 queries each). After normalisation, every remaining Latin token is one the language-ID step judged English, so that column is 100% by construction.

Audit measure	Value
Code-mixed queries identical to the English query	13.3%
Code-mixed queries with no romanised-Tamil token	22.0%
Mean romanised-Tamil tokens per code-mixed query	1.74
Mean romanised-Tamil tokens per romanised query	4.30

Table 5.3: How much Tamil the dataset's code-mixed queries contain.

The audit gives two findings that shape how the results should be read:

1. **The romanised queries are genuinely Tamil.** They contain on average 4.3 tokens detected as romanised Tamil, and only 14% of their Latin tokens look English, mostly names such as *Colombia* or *London*.
2. **The "code-mixed" queries are mostly English.** On average only 1.74 tokens per code-mixed query are Tamil; 22.0% contain no Tamil token at all, and 13.3% are identical to the English query. This is a property of the LLM that generated the dataset, not of how people type Tanglish. Results for the code-mixed set therefore mostly measure English-to-Tamil cross-lingual retrieval, which is the main reason Module 3 will collect human-written Tanglish questions.

Transliteration also changes vocabulary overlap with the corpus. Of the Tamil-script tokens in native Tamil queries, 77.7% appear in the corpus vocabulary. After rule-based transliteration of romanised queries, 29.9% do, against 15.2% for ITRANS. Exact word matches remain rare even after normalisation, which is why character n-gram BM25 benefits most.

5.3 Main results: raw queries on the Tamil corpus

Query form	Retriever	R@1	R@5	R@10	MRR@10	MRR@10 95% CI	nDCG@10
Tamil script	BM25	0.764	0.872	0.891	0.811	[0.795, 0.825]	0.830
	BM25-char3	0.898	0.954	0.964	**0.923**	[0.912, 0.933]	0.933
	Dense-e5	0.862	0.956	0.974	0.903	[0.893, 0.913]	0.921
	Hybrid-RRF	0.860	0.937	0.960	0.895	[0.884, 0.907]	0.911
	Hybrid-RRF3	0.883	0.945	0.970	0.911	[0.900, 0.921]	0.925
English	BM25	0.062	0.075	0.078	0.067	[0.057, 0.078]	0.070
	BM25-char3	0.051	0.075	0.088	0.062	[0.053, 0.072]	0.068
	Dense-e5	0.711	0.892	0.937	**0.788**	[0.774, 0.803]	0.824
	Hybrid-RRF	0.567	0.889	0.936	0.710	[0.695, 0.726]	0.766
	Hybrid-RRF3	0.085	0.695	0.879	0.305	[0.295, 0.316]	0.442
Code-mixed	BM25	0.065	0.075	0.079	0.069	[0.059, 0.080]	0.072
	BM25-char3	0.052	0.078	0.088	0.063	[0.054, 0.073]	0.069
	Dense-e5	0.597	0.805	0.862	**0.687**	[0.670, 0.706]	0.729
	Hybrid-RRF	0.505	0.806	0.863	0.637	[0.622, 0.655]	0.693
	Hybrid-RRF3	0.085	0.616	0.801	0.278	[0.268, 0.290]	0.403
Romanised	BM25	0.051	0.063	0.068	0.056	[0.046, 0.065]	0.059
	BM25-char3	0.048	0.068	0.082	0.058	[0.048, 0.067]	0.064
	Dense-e5	0.093	0.212	0.294	0.143	[0.132, 0.157]	0.178
	Hybrid-RRF	0.129	0.247	0.321	**0.178**	[0.164, 0.192]	0.211
	Hybrid-RRF3	0.065	0.178	0.262	0.113	[0.103, 0.123]	0.148

Table 5.4: Raw queries on the Tamil corpus (n = 2,154 per form). Bold marks the best MRR@10 per form.

RQ1 and RQ2. With Tamil-script queries every retriever works well (MRR@10 between 0.811 and 0.923). English queries are retrieved well only by the dense model (0.788), because BM25 cannot match English words to Tamil text. Code-mixed queries behave like English queries (dense 0.687). **Romanised queries fail everywhere:** dense retrieval

reaches MRR@10 = 0.143 and Recall@10 = 0.294, so for 7 in 10 romanised questions the correct document is not in the top 10; the best raw configuration (BM25 + dense fusion) reaches only 0.178. The multilingual embedding model, although trained on about 100 languages, does not place romanised Tamil near Tamil script.

Why character 3-grams beat word BM25 on native Tamil (0.923 against 0.811). Tamil is agglutinative: case markers, postpositions and clitics attach to the stem, so one noun appears in many surface forms (மோதல்மோதலுக்கு, மோதலின். No Tamil stemmer or lemmatiser was used, so word BM25 counts these as unrelated terms, and a query word matches a document only if the suffix is identical. Character 3-grams share the stem's grams across all these forms, which acts as an approximate, language-independent stemmer. Only 77.7% of native-query Tamil tokens occur verbatim in the corpus vocabulary (Section 5.2), which is the gap the 3-grams close.

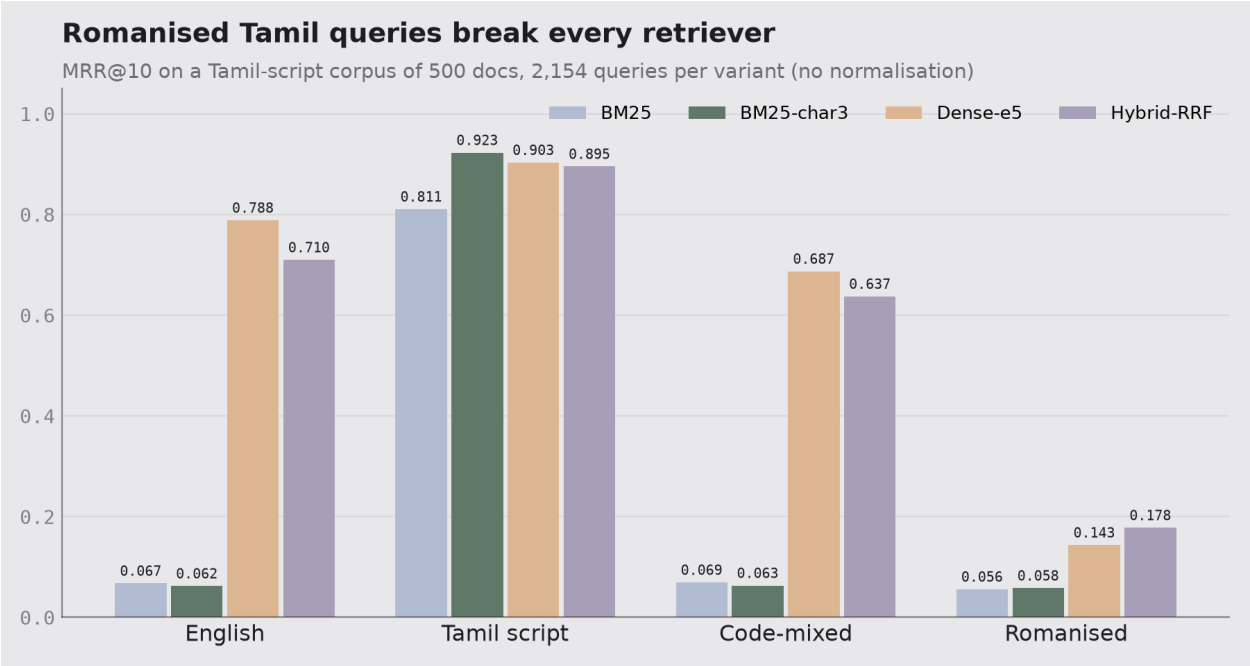


Figure 5.1: MRR@10 for each query form and retriever on the Tamil corpus, without normalisation.

5.4 Effect of query normalisation

Query form	Retriever	R@1	R@5	R@10	MRR@10	MRR@10 95% CI	nDCG@10
Romanised	BM25	0.051	0.063	0.068	0.056	[0.046, 0.065]	0.059
	BM25-char3	0.048	0.068	0.082	0.058	[0.048, 0.067]	0.064
	Dense-e5	0.093	0.212	0.294	0.143	[0.132, 0.157]	0.178
	Hybrid-RRF	0.129	0.247	0.321	**0.178**	[0.164, 0.192]	0.211
	Hybrid-RRF3	0.065	0.178	0.262	0.113	[0.103, 0.123]	0.148
Romanised + ITRANS	BM25	0.174	0.257	0.290	0.210	[0.194, 0.226]	0.229
	BM25-char3	0.475	0.638	0.705	**0.547**	[0.529, 0.567]	0.585
	Dense-e5	0.253	0.434	0.518	0.328	[0.310, 0.346]	0.373
	Hybrid-RRF	0.305	0.487	0.592	0.386	[0.366, 0.404]	0.435
	Hybrid-RRF3	0.417	0.631	0.718	0.506	[0.486, 0.522]	0.556
Romanised + rules	BM25	0.279	0.405	0.449	0.333	[0.314, 0.351]	0.361
	BM25-char3	0.623	0.766	0.812	**0.684**	[0.666, 0.702]	0.715

Query form	Retriever	R@1	R@5	R@10	MRR@10	MRR@10 95% CI	nDCG@10
	Dense-e5	0.443	0.657	0.746	0.535	[0.515, 0.554]	0.585
	Hybrid-RRF	0.448	0.648	0.742	0.537	[0.518, 0.556]	0.586
	Hybrid-RRF3	0.539	0.743	0.842	0.627	[0.609, 0.646]	0.678
Code-mixed	BM25	0.065	0.075	0.079	0.069	[0.059, 0.080]	0.072
	BM25-char3	0.052	0.078	0.088	0.063	[0.054, 0.073]	0.069
	Dense-e5	0.597	0.805	0.862	**0.687**	[0.670, 0.706]	0.729
	Hybrid-RRF	0.505	0.806	0.863	0.637	[0.622, 0.655]	0.693
	Hybrid-RRF3	0.085	0.616	0.801	0.278	[0.268, 0.290]	0.403
Code-mixed + ITRANS	BM25	0.080	0.109	0.116	0.093	[0.082, 0.105]	0.099
	BM25-char3	0.145	0.267	0.322	0.197	[0.184, 0.213]	0.227
	Dense-e5	0.570	0.793	0.859	**0.666**	[0.649, 0.685]	0.713
	Hybrid-RRF	0.431	0.777	0.855	0.585	[0.569, 0.604]	0.651
	Hybrid-RRF3	0.294	0.584	0.703	0.412	[0.395, 0.431]	0.481
Code-mixed + rules	BM25	0.097	0.128	0.135	0.111	[0.098, 0.124]	0.117
	BM25-char3	0.181	0.312	0.369	0.238	[0.223, 0.254]	0.269
	Dense-e5	0.597	0.816	0.880	**0.692**	[0.676, 0.710]	0.738
	Hybrid-RRF	0.422	0.745	0.841	0.568	[0.552, 0.586]	0.634
	Hybrid-RRF3	0.304	0.590	0.704	0.420	[0.402, 0.438]	0.488

Table 5.5: Effect of query normalisation on the Tamil corpus. Bold marks the best MRR@10 per query set.

RQ3. For romanised queries, normalisation is the largest effect in the study:

- Rules + character-3-gram BM25 raises MRR@10 from 0.058 to **0.684** and Recall@10 from 0.082 to **0.812**.
- Rules + dense raises MRR@10 from 0.143 to 0.535.
- Rules + RRF3 gives the best Recall@10 of all (0.842), useful when the generator reads the top 10, although its MRR@10 (0.627) is below character-3-gram BM25 alone (0.684).
- The Tanglish rules beat ITRANS on every retriever (character-3-gram BM25: 0.684 against 0.547 MRR@10).
- A gap remains: the best romanised result (0.684) is still well below Tamil script (0.923), because Tanglish spelling loses distinctions such as ல/ள, ன/ண/ந and த/ட.

For code-mixed queries, normalisation helps the lexical retrievers (BM25 0.069 to 0.111; RRF3 0.278 to 0.420) but **does not change dense retrieval** (0.687 to 0.692, p = 0.45). These queries are mostly English (Section 5.2), so there is little Tamil to transliterate, and the dense model already handles the English part.

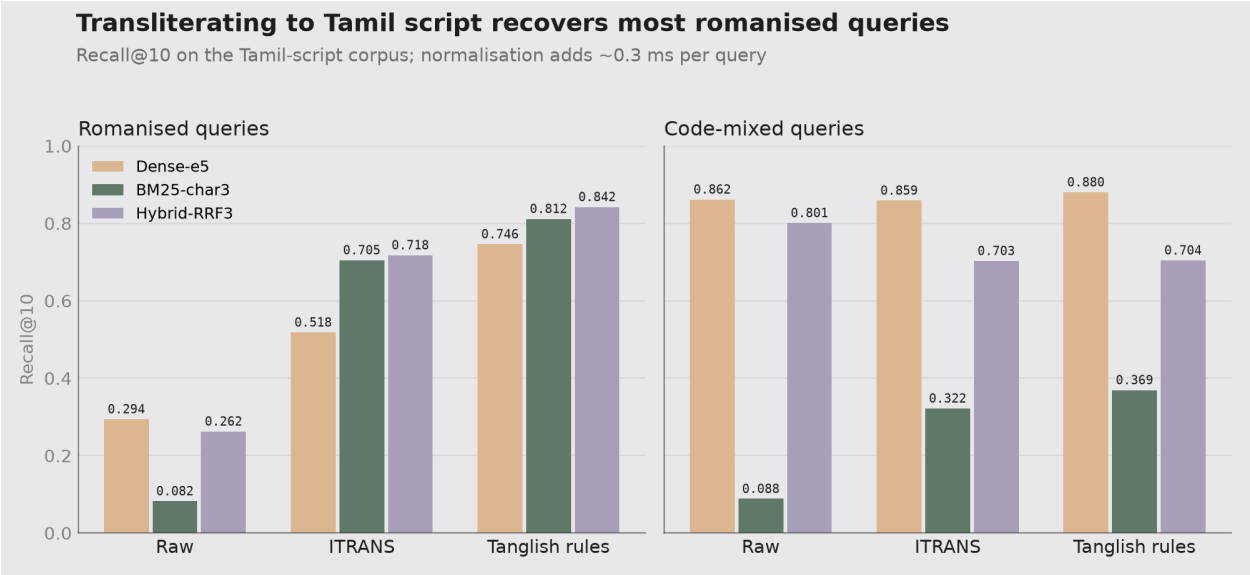


Figure 5.2: Recall@10 for raw, ITRANS-normalised and rule-normalised romanised and code-mixed queries.

5.5 Statistical significance

Comparison	Retriever(s)	MRR@10 before	MRR@10 after	Change	p (query bootstrap)	p (cluster bootstrap)	Change, 95% CI (cluster)
Romanised → Romanised + rules	Dense	0.143	0.535	+0.391	< 0.001	< 0.001	[+0.365, +0.417]
Romanised → Romanised + rules	BM25 char-3	0.058	0.684	+0.626	< 0.001	< 0.001	[+0.596, +0.655]
Romanised → Romanised + rules	Hybrid-RRF3	0.113	0.627	+0.514	< 0.001	< 0.001	[+0.485, +0.542]
Romanised + ITRANS → Romanised + rules	Dense	0.328	0.535	+0.207	< 0.001	< 0.001	[+0.184, +0.230]
Romanised + ITRANS → Romanised + rules	BM25 char-3	0.547	0.684	+0.137	< 0.001	< 0.001	[+0.113, +0.161]
Code-mixed → Code-mixed + rules	Dense	0.687	0.692	+0.005	0.446	0.573	[-0.013, +0.022]
Code-mixed → Code-mixed + rules	BM25	0.069	0.111	+0.042	< 0.001	< 0.001	[+0.024, +0.062]
Code-mixed → Code-mixed + rules	Hybrid-RRF3	0.278	0.420	+0.142	< 0.001	< 0.001	[+0.115, +0.167]
Tamil script: fusion	Dense → Hybrid-RRF	0.903	0.895	-0.008	0.168	0.287	[-0.023, +0.007]
Tamil script: fusion	Dense → Hybrid-RRF (naive, v1)	0.903	0.884	-0.019	0.003	0.032	[-0.036, -0.002]
English: fusion	Dense → Hybrid-RRF	0.788	0.710	-0.079	< 0.001	< 0.001	[-0.090, -0.068]
English: fusion	Dense → Hybrid-RRF (naive, v1)	0.788	0.187	-0.601	< 0.001	< 0.001	[-0.634, -0.567]
Code-mixed: fusion	Dense → Hybrid-RRF	0.687	0.637	-0.050	< 0.001	< 0.001	[-0.062, -0.038]
Code-mixed: fusion	Dense → Hybrid-RRF (naive, v1)	0.687	0.181	-0.506	< 0.001	< 0.001	[-0.540, -0.471]

Comparison	Retriever(s)	MRR@10 before	MRR@10 after	Change	p (query bootstrap)	p (cluster bootstrap)	Change, 95% CI (cluster)
Romanised: fusion	Dense → Hybrid-RRF	0.143	0.178	+0.034	< 0.001	< 0.001	[+0.023, +0.046]
Romanised: fusion	Dense → Hybrid-RRF (naive, v1)	0.143	0.092	-0.052	< 0.001	< 0.001	[-0.075, -0.030]
Romanised + rules: fusion	Dense → Hybrid-RRF	0.535	0.537	+0.002	0.803	0.856	[-0.023, +0.026]
Romanised + rules: fusion	Dense → Hybrid-RRF (naive, v1)	0.535	0.441	-0.094	< 0.001	< 0.001	[-0.126, -0.062]
Romanised + rules: fusion	BM25 char-3 → Hybrid-RRF3	0.684	0.627	-0.057	< 0.001	< 0.001	[-0.080, -0.034]

Table 5.6: Paired significance tests on the Tamil corpus (2,000 resamples each). The query bootstrap resamples queries; the cluster bootstrap resamples the 500 source documents with all their queries. Changes are computed from unrounded values, so a change can differ by 0.001 from the difference of the rounded MRR@10 columns. "Naive, v1" is the earlier fusion that included zero-score BM25 documents.

Every improvement from the Tanglish rules for romanised queries is significant ($p < 0.001$), as is the rules-over-ITRANS improvement, under both the query-level and the stricter document-cluster bootstrap. The dense change for code-mixed queries is not significant ($p = 0.45$; cluster $p = 0.57$). The cross-script losses caused by equal-weight fusion remain significant under both tests; the small loss for Tamil-script queries does not ($p = 0.17$; cluster $p = 0.29$). The cluster bootstrap widens the MRR@10 confidence intervals by about 1.6 times (for example dense on English queries: query-level [0.774, 0.803], cluster [0.766, 0.810]), but no conclusion in this report changes.

5.6 Hybrid fusion can hurt

Hybrid BM25 + dense retrieval with RRF is widely recommended as a robust default. With equal weights and no script check, it is not robust across scripts. When the query and the documents are in different scripts (English or code-mixed queries against Tamil text), word BM25 matches nothing for most queries, and for the rest it matches only incidental tokens: the commonest English words that also occur in the Tamil documents are *of*, *tamil*, *sri*, *from*, *news* and *a*. RRF still gives that list the same weight as the dense list, so the correct document ranked high by the dense model is pushed down. With zero-score documents correctly excluded:

- English queries: dense 0.788, Hybrid-RRF 0.710 (−0.079, $p < 0.001$; cluster $p < 0.001$).
- Code-mixed queries: dense 0.687, Hybrid-RRF 0.637 (−0.050, $p < 0.001$; cluster $p < 0.001$).
- Adding the character-3-gram list makes it much worse (English 0.305, code-mixed 0.278), because character 3-grams of English words also match grams of the digits and stray Latin words inside Tamil documents (median 10 matching documents per English query), and these weak matches displace the dense result.
- For Tamil-script queries, fusion is slightly below dense alone (0.903 to 0.895, not significant) and below character-3-gram BM25 alone (0.923).

The bug and what it changed. The earlier, naive run fed zero-score documents into RRF in index order, which turned "BM25 found nothing" into "BM25 strongly recommends documents 0 to 99". That inflated the harm seven- to tenfold: English 0.788 → 0.187 (−0.601) and code-mixed 0.687 → 0.181 (−0.506). The corrected losses are much smaller, but they are

still significant, so the conclusion that equal-weight hybrid fusion hurts across scripts stands, at its corrected size.

Fusion helps when the lexical list carries real signal. For raw romanised queries the few word-BM25 matches are mostly names and numbers (*venugopal*, *amir*, *2000*, *1984*), and fusion lifts MRR@10 from 0.143 to 0.178 ($p < 0.001$). After rule normalisation, RRF3 gives the best Recall@10 (0.842). **Practical implication:** a code-mixed RAG system should choose or weight retrievers according to the query's script, and must drop zero-score lexical results before fusing, rather than fuse blindly.

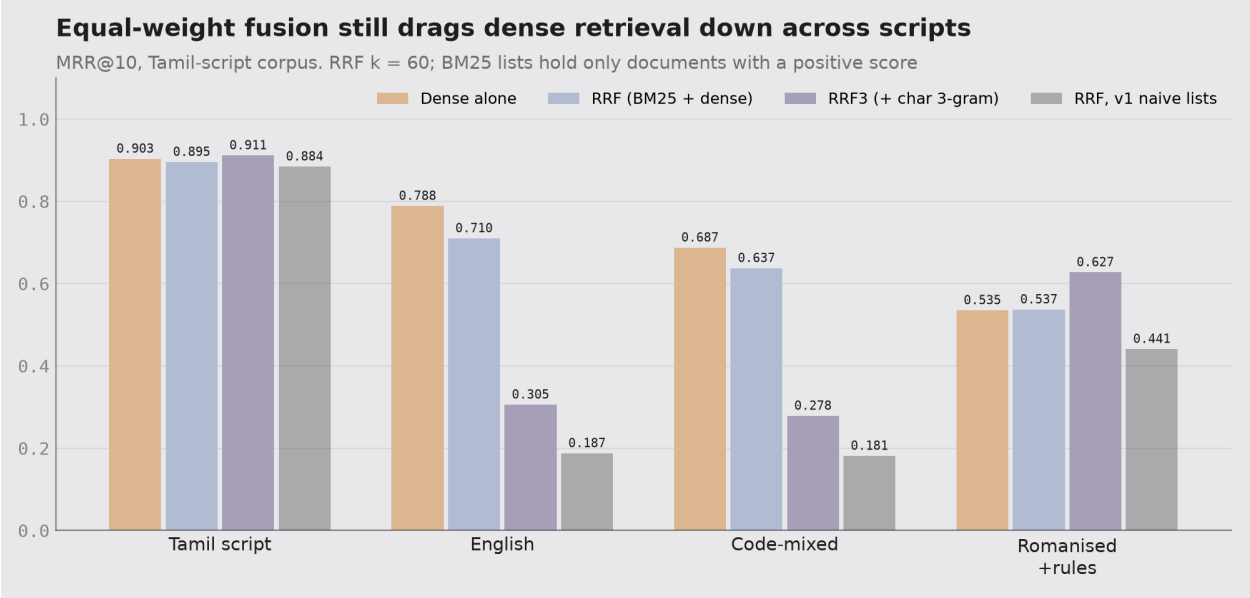


Figure 5.3: MRR@10 of dense retrieval alone against the corrected RRF and RRF3 hybrids, with the earlier naive RRF (zero-score documents not removed) for contrast.

5.7 Secondary setting: English corpus

Query form	Retriever	R@1	R@5	R@10	MRR@10	MRR@10 95% CI	nDCG@10
English	BM25	0.838	0.918	0.929	0.870	[0.857, 0.883]	0.885
	BM25-char3	0.845	0.914	0.937	0.876	[0.863, 0.888]	0.891
	Dense-e5	0.915	0.972	0.984	**0.939**	[0.929, 0.947]	0.950
	Hybrid-RRF	0.894	0.950	0.964	0.920	[0.909, 0.930]	0.930
	Hybrid-RRF3	0.890	0.950	0.964	0.916	[0.905, 0.926]	0.927
Tamil script	BM25	0.058	0.071	0.076	0.064	[0.054, 0.074]	0.067
	BM25-char3	0.036	0.067	0.084	0.050	[0.041, 0.059]	0.058
	Dense-e5	0.664	0.850	0.896	**0.742**	[0.726, 0.760]	0.779
	Hybrid-RRF	0.643	0.846	0.896	0.728	[0.713, 0.746]	0.769
	Hybrid-RRF3	0.089	0.132	0.212	0.113	[0.101, 0.127]	0.135
Code-mixed	BM25	0.833	0.905	0.922	0.865	[0.851, 0.878]	0.879
	BM25-char3	0.820	0.899	0.916	0.855	[0.840, 0.868]	0.870
	Dense-e5	0.838	0.923	0.942	0.875	[0.861, 0.888]	0.891
	Hybrid-RRF	0.870	0.938	0.957	0.900	[0.888, 0.910]	0.914
	Hybrid-RRF3	0.875	0.935	0.952	0.901	[0.889, 0.912]	0.913
Romanised	BM25	0.493	0.560	0.571	0.522	[0.503, 0.541]	0.534

Query form	Retriever	R@1	R@5	R@10	MRR@10	MRR@10 95% CI	nDCG@10
	BM25-char3	0.392	0.518	0.560	0.445	[0.426, 0.465]	0.472
	Dense-e5	0.278	0.412	0.466	0.333	[0.314, 0.353]	0.364
	Hybrid-RRF	0.506	0.595	0.619	0.544	[0.524, 0.564]	0.562
	Hybrid-RRF3	0.520	0.605	0.629	**0.557**	[0.537, 0.577]	0.575

Table 5.7: Raw queries on the English-translation corpus. Bold marks the best MRR@10 per form.

When the documents are English translations, English and code-mixed queries are easy (dense 0.939; RRF3 0.901 for code-mixed). Tamil-script queries are handled only by the dense model (0.742; word-BM25 + dense fusion 0.728). Romanised queries are again the hardest form (best 0.557 with RRF3); lexical retrievers do comparatively well on them because romanised queries keep English named entities. Whatever the document language, romanised Tamil is the weakest query form.

5.8 Latency

Stage	Median (ms)	95th percentile (ms)
Tanglish rule normalisation	0.34	0.48
BM25 (word) scoring + top-k	0.67	0.98
BM25 (char 3-gram) scoring + top-k	4.88	8.59
Dense query encoding (mE5-small, CPU)	32.11	39.26
Dense search (500 × 384 dot product)	0.14	0.18
RRF fusion of 3 lists	0.19	0.23
Full rules + RRF3 pipeline (sum of medians)	**38.3**	

Table 5.8: Per-query latency, one query at a time on 2 CPU cores (200 random romanised queries; index build and document encoding excluded).

The normaliser costs about 0.3 ms per query, which is negligible. Dense query encoding dominates (about 32 ms on 2 CPU cores). The best romanised pipeline (rules + character-3-gram BM25) needs about 5 ms per query with no neural model at all, and the full rules + RRF3 pipeline about 38 ms (sum of stage medians). All are well within interactive limits.

5.9 Example outputs

The following are real query-to-retrieval outputs of the dense retriever on the Tamil corpus. “Rank of gold” is the position of the correct document in the top 10; a dash means it is not in the top 10.

Example 1. Gold document (English translation, shortened): “A guerrilla attack on military personnel in the municipality of Aragita, in the Arawka region of Colombia, has killed 5 soldiers.”

Form	Query as given to the retriever	Rank of gold	Top-1 retrieved document (Tamil, shortened)
English	Colombia military soldiers killed in attack	1	✓ correct: கொலம்பியா நாட்டின் அராவ்கா துறைக்கு உட்பட்ட அராகிட்டா நகராட்சி பகுதியி…

Form	Query as given to the retriever	Rank of gold	Top-1 retrieved document (Tamil, shortened)
Tamil script	தாக்குதலில் கொலம்பியா ராணுவ வீரர்கள் உயிரிழப்பு	1	✓ correct: கொலம்பியா நாட்டின் அராவ்கா துறைக்கு உட்பட்ட அராகிட்டா நகராட்சி பகுதியி…
Roman-ised	thakkudhalil Colombia raanuva thiraigal uyirizhappu	—	✗ wrong: பஞ்சாலை தொழிலாளர்களின் வாழ்வியலையும்உடுமலை வட்டாரப் பகுதிகளில் முன்ப…
Roman-ised + rules	தக்குதலில் Colombia ரானுவ திரைகல் உயிரிழப்பு	1	✓ correct: கொலம்பியா நாட்டின் அராவ்கா துறைக்கு உட்பட்ட அராகிட்டா நகராட்சி பகுதியி…
Code-mixed	Colombia military soldiers killed in attack	1	✓ correct: கொலம்பியா நாட்டின் அராவ்கா துறைக்கு உட்பட்ட அராகிட்டா நகராட்சி பகுதியி…
Code-mixed + rules	Colombia military soldiers killed in attack	1	✓ correct: கொலம்பியா நாட்டின் அராவ்கா துறைக்கு உட்பட்ட அராகிட்டா நகராட்சி பகுதியி…

Example 2. Gold document (English translation, shortened): “A member of the Northern Province ஜி.ரி.லிங்கநாதன்has demanded that the length of the uniform of the schoolgirls of the Northern Province b…”

Form	Query as given to the retriever	Rank of gold	Top-1 retrieved document (Tamil, shortened)
English	Northern Province school uniform length request	1	✓ correct: வடக்கு மாகாண பாடசாலை மாணவிகளின் சீருடையின் நீளத்தை அதிகரிக்க வேண்டும்..
Tamil script	வடக்கு மாகாண பாடசாலை சீருடை நீளம் கோரிக்கை	1	✓ correct: வடக்கு மாகாண பாடசாலை மாணவிகளின் சீருடையின் நீளத்தை அதிகரிக்க வேண்டும்..
Roman-ised	Vadakku maagaana paadasaalai seerudai neelam korikkai	—	✗ wrong: “பின்னே என்ன சார்? ஒருத்தரை வெறுக்கிறோம்னா அதை வெளிப்படையா சொல்லலாமே? …
Roman-ised + rules	வடக்கு மாகான பாடசாலை சீருடை நீலம் கொரிக்கை	1	✓ correct: வடக்கு மாகாண பாடசாலை மாணவிகளின் சீருடையின் நீளத்தை அதிகரிக்க வேண்டும்..
Code-mixed	Northern province school uniform length increase request	1	✓ correct: வடக்கு மாகாண பாடசாலை மாணவிகளின் சீருடையின் நீளத்தை அதிகரிக்க வேண்டும்..
Code-mixed + rules	Northern province school uniform length increase request	1	✓ correct: வடக்கு மாகாண பாடசாலை மாணவிகளின் சீருடையின் நீளத்தை அதிகரிக்க வேண்டும்..

Example 3. Gold document (English translation, shortened): “New Delhi: The Triple Talaq (Prohibition) Bill is being introduced in the Rajya Sabha today (December 31), following which the BJP-Congress …”

Form	Query as given to the retriever	Rank of gold	Top-1 retrieved document (Tamil, shortened)
English	Triple Talaq Bill Rajya Sabha introduction	3	✗ wrong: மேற்கண்ட தகவல்கள்2006 ஜூன் மல்லிகையில் மேமன்கவி அவர்கள் எழுதிய, “எழுத…
Tamil script	முத்தலாக் தடை மசோதா ராஜ்யசபா	1	✓ correct: புதுடில்லி : ‘முத்தலாக்தடை மசோதா, ராஜ்யசபாவில் இன்று(டிச.,31) தாக்கல…
Romanised	Muthalaq thadai masoda Rajyasabha	—	✗ wrong: அழகிய மணிமுடியும்தாமரை மலர் போல சிவந்த திருக்கரங்களும்ஒளிவீசும் வே…
Romanised + rules	முதலக் தடை மசொட ரஜ்யசப	—	✗ wrong: எந்தக் கட்சி ஆட்சியில் இருந்தாலும் ’நான் அடிப்பதுபோல் அடிக்கிறேன்நீ…
Code-mixed	Triple Talaq bill Rajya Sabha la eppo	4	✗ wrong: எந்தக் கட்சி ஆட்சியில் இருந்தாலும் ’நான் அடிப்பதுபோல் அடிக்கிறேன்நீ…
Code-mixed + rules	Triple டலக் bill ரஜ்ய Sabha ல எப்பொ	—	✗ wrong: எந்தக் கட்சி ஆட்சியில் இருந்தாலும் ’நான் அடிப்பதுபோல் அடிக்கிறேன்நீ…

Table 5.9: Example outputs of the dense retriever (multilingual-e5-small) on the Tamil corpus.

Example 3 shows the limits of rule-based transliteration. *Muthalaq* (முத்தலாக் “triple talaq”) becomes முதலக் and *masoda* becomes மசொட instead of மசோதா (“bill”), because the romanised spelling marks neither the doubled *t*, the long *o* nor the final long *a*. Neither the dense model nor the rules recover this query, which points to character-level matching and better transliteration as Module 3 work. The code-mixed version shows a second failure: the language-ID step keeps *Triple*, *bill* and *Sabha* as English but sends *Talaq* and *Rajya* through the Tamil rules, giving “Triple டலக்bill ரஜ்ய Sabha”, which splits one named entity across two scripts.

5.10 Discussion

What the results mean for a RAG builder. A Tamil RAG system that passes Tanglish queries straight to a multilingual retriever misses the correct document for most romanised questions. A transliteration step of a few dozen rules, costing under a millisecond, recovers most of the loss, and a character n-gram index makes it robust to the remaining spelling errors and to Tamil’s agglutinative suffixes, which no stemmer handled here. Hybrid fusion should be gated by script, and zero-score lexical results must never enter the fusion.

Why dense models fail on romanised Tamil. Multilingual embedding models are trained largely on native-script text. Romanised Tamil is split into English-like sub-words with no learned link to the Tamil-script forms. The model does better on code-mixed queries only because those queries are mostly English.

Threats to validity.

1. *Synthetic queries.* Every query was generated by an LLM. Real Tanglish has more Tamil words, more spelling variation and more colloquial grammar. Module 3 tests on human-written queries.
2. *An easy retrieval setting.* Each query was generated from its own document, so lexical overlap with the gold document is higher than in real use. The 500-document corpus is small, and Tamil-script scores are near ceiling.

3. *One relevant document per query.* Other documents could also answer some queries but are counted as wrong, so scores are a lower bound.
4. *One small dense model.* Larger models (BGE-M3, multilingual-e5-base or -large, LaBSE) may behave differently; they are scheduled for Module 3.
5. *Rules written by one author.* The rules were not tuned on test data, but they reflect one person's view of Tanglish spelling. A held-out human set will check that they generalise.
6. *Language ID breaks named entities.* The frequency-threshold language ID (wordfreq Zipf ≥ 3.0) knows English words but not Indian names. Indian named entities written in Roman script, such as *Talaq* or *Rajya*, are judged non-English and transliterated, while their English neighbours are kept, so one entity can end up half in each script (Section 5.9, Example 3). An entity-aware language ID is planned (Section 6.2).
7. *Queries cluster within documents.* The 2,154 queries come from 500 documents (about 4.3 each), so they are not independent. Query-level bootstrap intervals are therefore too narrow. Re-running every key test with a document-cluster bootstrap widens the intervals by about 1.6 times but leaves every conclusion unchanged (Section 5.5).
8. *An earlier fusion bug.* The first run fed zero-score BM25 documents into RRF. It is fixed, and the naive numbers are kept only for comparison (Sections 4.4 and 5.6).

Chapter 6: Conclusion and Future Work

6.1 Conclusion

Modules 1 and 2 defined the problem of code-mixed Tamil–English retrieval for RAG, surveyed 19 related works including the FIRE mixed-script IR tracks and identified a narrower but clear gap, designed a script-aware retrieval pipeline, and implemented and evaluated it on 2,154 queries in four forms with five retrievers and two normalisation methods on a CPU. The main findings:

1. **Romanised Tamil queries break standard retrieval.** Dense retrieval reaches only MRR@10 = 0.143 and the best raw configuration 0.178, against 0.923 for Tamil script.
2. **Transliteration to Tamil script recovers most of the loss cheaply.** Tanglish rules with character-3-gram BM25 reach MRR@10 = 0.684 and Recall@10 = 0.812 at about 0.3 ms normalisation cost, and significantly beat the generic ITRANS scheme.
3. **Naive (equal-weight, ungated) hybrid fusion is harmful across scripts.** With zero-score documents correctly excluded, RRF still cuts dense MRR@10 from 0.788 to 0.710 for English queries on Tamil documents (0.305 with a third, character-3-gram list). An earlier run that fused zero-score documents overstated this as 0.187.
4. **The public code-mixed query set is English-heavy.** 22% of its code-mixed queries contain no Tamil word, so human-written Tanglish data is needed.

6.2 Future modules

Module 3: Human Tanglish questions and Tamil-aware chunking.

- Collect about 500 human-written Tanglish questions from Tamil-speaking students over a longer document collection (for example college circulars, scholarship notices and government notices), with relevance judgements.
- Implement a Tamil-aware chunker that splits at Tamil sentence and clause boundaries and never inside a grapheme cluster, and compare it with the ChunkLab chunkers (fixed-size, recursive, sentence, semantic) at several chunk sizes.
- Make a **script-gated router** the headline system: detect the query's script mix and send it to the retriever that works for that form (dense for English and code-mixed, rules + character-3-gram BM25 for romanised Tamil), fusing only when every list is informative.
- Add baselines: **IndicXlit** neural transliteration [22] in place of the rules, and an **LLM query rewriter** that converts a Tanglish query into Tamil script or English before retrieval.
- Use **longer real documents**, such as Tamil Nadu scholarship notices and Government Orders (G.O.s), so that chunking actually matters.
- Try a **phonetic-skeleton BM25** that indexes consonant skeletons with vowel signs removed and merges ல/ள/ழ, ன/ண/ந and த/ட, to absorb what Tanglish spelling cannot encode (a FIRE 2015 system already removed vowel signs before character n-gram indexing [20]).
- Build an **entity-safe language ID** that keeps Indian named entities (*Rajya Sabha*, *Triple Talaq*) whole, using a gazetteer and capitalisation, instead of a pure English-frequency threshold.

- Add BGE-M3 and LaBSE.

Module 4: Generation-side RAG evaluation. Connect an LLM generator and measure answer correctness and faithfulness for each query form, to test whether retrieval gains carry through to answers.

Module 5: Paper and release. Write up the full factorial study for a workshop or conference (for example FIRE or a code-switching workshop) and release the benchmark and code.

6.3 Timeline

Period	Work
Nov 2026	Module 2 review; design the human query collection protocol
Dec 2026 – Jan 2027	Module 3: collect ~500 Tanglish questions; Tamil-aware chunker; larger embedding models
Feb – Mar 2027	Module 4: LLM generation and answer-quality evaluation
Apr – May 2027	Module 5: paper, thesis write-up, public release

Table 6.1: Planned timeline.

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